ViGALT: Visual Gap-Aware Alt Text Generation for Blind and Low-Vision E-Commerce Shoppers

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***Abstract*—Product images on e-commerce platforms often carry alt text that repeats the product title or is missing altogether, leaving blind and low-vision (BLV) shoppers without the visual details they need to decide whether to buy. Context-aware description systems already use webpage context to improve image descriptions, but they are not designed around what a shopper still needs to know after the page text has been read. We present ViGALT (Visual Gap-Aware Alt Text Generation), a multi-stage pipeline that treats the product page as information the user already has and writes alt text that fills the remaining visual gap. ViGALT combines category routing, deterministic extraction of product attributes from the page, generation with a vision language model, decomposition into atomic claims, three generation-time validators for redundancy, accuracy, and completeness, and a constrained fuser. On a paired set of 30 Amazon.fr products, with the same model backend for both systems, ViGALT raised claim-level relevancy from 48.12 to 90.10 percent over a reproduced ASSETS 2024 context-aware baseline, a gain that remains significant after correction for multiple comparisons. Objectivity and relevant novelty improved in the same direction. A fusion-stage ablation shows that the completeness and accuracy validators make distinct, traceable contributions to the final text.**

***Index Terms*—accessibility, alternative text, blind and low vision, e-commerce, vision language models, large language models, image description**

I. Introduction

Online shopping relies on images to convey shape, color, material, and fit, which creates a barrier for Blind and Low-Vision (BLV) users who browse with screen readers. Alternative text (alt text) is "descriptive text that conveys the meaning of an image in digital content" [1]; it is the main channel through which a BLV shopper can access what a product photograph shows.

In practice, e-commerce alt text is frequently insufficient. Many sites duplicate the product title, paste fragments of the description, or, at best, run a generic image captioner. Because large marketplaces let third-party sellers supply their own metadata, alt-text quality also varies widely between listings. This matters because BLV shoppers specifically want appearance attributes such as color, logo, shape, and size [2], and generic alt text does not support the decision to buy or not.

Vision Language Models (VLMs) can now describe images fluently, but good alt text depends on the image's context and purpose [3]: the same image may need a completely different description depending on where it appears and what the user is trying to do [4]. W3C guidance makes the same point and adds that if nearby text already conveys the visual information, repeating it in the text alternative is redundant [5], [6]. On a product page, the title and attribute list already tell the user that an item is, say, a black women's sweater. The useful question is therefore not "what is in this image?" but "what visual information is still missing from this page?"

We introduce ViGALT (Visual Gap-Aware ALT Text Generation), a multi-stage pipeline that answers that question. ViGALT extracts the product page's Document Object Model (DOM) into structured context with a deterministic parser; all subsequent generation, validation, and rewriting steps are performed by Large Language Models (LLMs). Our contributions are:

- a gap-aware generation pipeline that decomposes a candidate description into atomic claims and checks them with three parallel validators (redundancy, accuracy, completeness) before a constrained fuser writes the final alt text;
- a paired, claim-level comparison against a reproduced context-aware baseline [7] under a controlled model backend, with Wilcoxon signed-rank tests and Holm correction;
- a fusion-stage ablation that isolates what each validator contributes to the final text.

II. Related Work

Image-only description and evaluation. Reference-free metrics such as CLIPScore score the compatibility between an image and a caption [8], and captioning datasets for BLV users, such as VizWiz-Captions, target general scene description [9]. None of these reason about what the surrounding page already tells the reader. Kreiss et al. showed that referenceless metrics align poorly with accessibility judgments precisely because context is missing, and called for context-aware evaluation [10].

Context-aware generation. Stangl et al. found that BLV users' information wants depend on both the image source and the user's goal [4]. Mohanbabu and Pavel (ASSETS 2024) extract webpage context and combine it with the image to produce context-aware descriptions; in their BLV user study, these were generally preferred to context-free descriptions [7]. We use this system, hereafter ASSETS24, as our baseline. It is, however, general-purpose rather than tuned to product evaluation. In our reproduction on Amazon.fr listings, it often produced details that were visually correct but irrelevant to the purchase, such as a model's jeans, necklaces, or handbag. Such

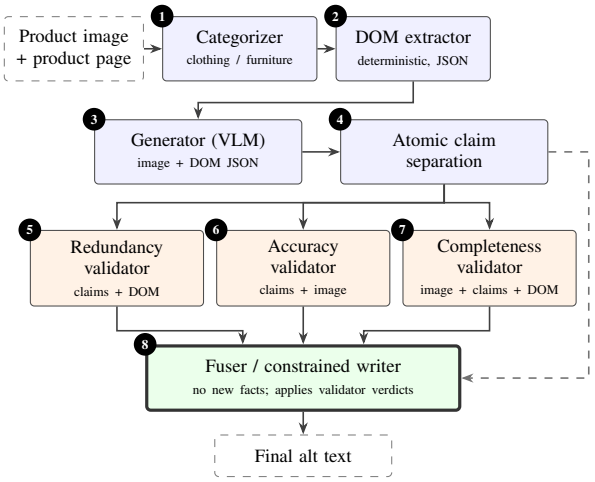


Fig. 1. ViGALT generation pipeline. Validators 5–7 run in parallel on the same atomic claims and act as generation-time safeguards; they are distinct from the post-hoc evaluation metrics of Section IV. The fuser also receives the candidate text and claims (dashed).

details inflate apparent novelty without helping the shopper. ViGALT therefore treats relevance to the target product and redundancy with the DOM as explicit objectives.

III. Methodology

Fig. 1 shows the ViGALT generation pipeline, which currently targets two product families: clothing and furniture.

1) *Categorizer.* Assigns the product to clothing or furniture so that later prompts focus on category-appropriate attributes.
2) *DOM extractor.* A deterministic script extracts product information from the page into structured JSON (e.g., color, material, sleeve type, neckline, and fit for clothing; analogous attributes for furniture). This JSON represents what the user can already learn from the page.
3) *Generator.* The image and DOM JSON are passed to a VLM (GPT-5.6 Luna), instructed to describe visually meaningful details not already covered by the DOM.
4) *Atomic claim separation.* The candidate is decomposed into independently checkable semantic claims, in the spirit of atomic-fact evaluation [11], so that validation operates on single facts rather than multi-fact sentences.
5) *Redundancy validator* (claims + DOM). Labels each claim `NOVEL`, `UNAVOIDABLE_REPETITION` (repetition needed to anchor a new visual relation, e.g., "the hoodie" in "the hoodie has a small logo on the left chest"), or `AVOIDABLE_REDUNDANCY`.
6) *Accuracy validator* (claims + image). Flags claims not supported by the image and may correct or delete them, reducing hallucination.
7) *Completeness validator* (image + claims + DOM). Lists important visible details absent from both the candidate and the DOM.
8) *Fuser.* A constrained writer that removes or corrects rejected content, incorporates validated missing details, and produces concise final alt text. It introduces no new facts of its own.

IV. Results

A. Experimental Setup

We collected 30 product images from Amazon.fr (15 clothing, 15 furniture) covering children's, sports, and women's garments with prints, logos, and characters, and beds, tables, mirrors, and sofas of varied shape, material, and staging. We ran the public ASSETS24 code [7] on the same images, replacing its deprecated GPT-4V backend with GPT-5.6 Luna, the model ViGALT uses. Differences can thus be attributed to pipeline design rather than model capability. Each final text from both systems is decomposed into atomic claims and judged by a stronger, identical judge (GPT-5.6 Terra). Scores are computed per image and macro-averaged, so each product counts equally.

Metrics. For an image with N atomic claims, the judge labels each claim; the per-image scores are

$$\mathrm{Obj} = N_{\mathrm{obj}}/N, \qquad \mathrm{Rel} = 100 \cdot \frac{N_{\mathrm{rel}} + 0.5N_{\mathrm{just}}}{N_{\mathrm{rel}} + N_{\mathrm{just}} + 2N_{\mathrm{viol}}}, \tag{1}$$

$$\mathrm{RN} = N_{\mathrm{nov}\wedge\mathrm{rel}}/N, \qquad \mathrm{AR} = N_{\mathrm{avoid}}/N, \tag{2}$$

$$\mathrm{Eff} = 100 \cdot N_{\mathrm{nov}\wedge\mathrm{rel}}/W, \tag{3}$$

where N_{obj} counts observable (non-subjective, non-marketing) claims; N_{rel}, N_{just}, and N_{viol} count on-product relevant (`R`), justified staging (`S`), and off-focus violation (`V`) claims assigned by the relevancy judge (same rubric as our automated evaluator); $N_{\mathrm{nov}\wedge\mathrm{rel}}$ counts claims that are `NOVEL` with respect to the DOM *and* relevant; N_{avoid} counts `AVOIDABLE_REDUNDANCY` claims; and W is the word count of the alt text. Relevancy uses fixed weights $w = 0.5$ on staging and $\lambda = 2$ on violations, so off-focus detail is penalized more strongly than in a plain relevant-claim fraction; clothing listings have $N_{\mathrm{just}} = 0$ by rubric. Per-image Rel is reported on a 0–100 scale; tables and figures macro-average $\mathrm{Rel}/100$ over the 30 products. Relevant Novelty (RN) prevents irrelevant scene detail from counting as useful novelty, and Information Efficiency (Eff) rewards concision, following W3C guidance [12]. Lower is better only for Avoidable Redundancy (AR). Paired differences are tested with the Wilcoxon signed-rank test [13], and p-values are Holm-corrected across the five metrics [14].

B. Comparison with the Baseline

Table I and Fig. 2 report the macro-averages and paired statistics. ViGALT scores higher on every metric, and its largest and only significant advantage after correction is in **Relevancy**: 90.10% versus 48.12% (+41.99 percentage points (pp); 95% CI [30.9, 53.2]; $p_{\mathrm{Holm}} = 4.6 \times 10^{-5}$; rank-biserial $r = 0.99$). ViGALT was more relevant on 25 of 30 products and less relevant on only one. This confirms statistically that the baseline's extra scene content is largely irrelevant to the purchase decision.

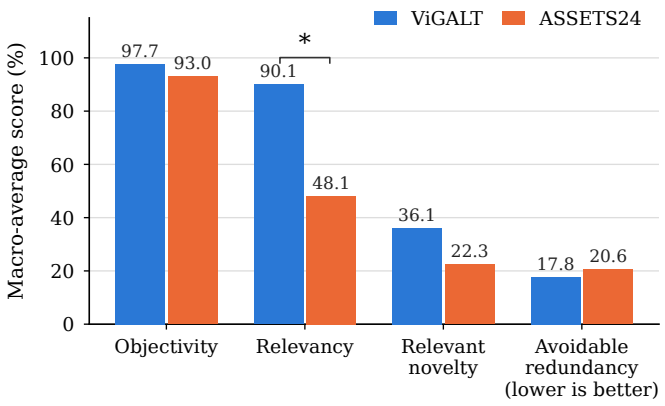


Fig. 2. Macro-averaged claim-level scores for ViGALT and ASSETS24 on the 30 paired products (Avoidable Redundancy: lower is better). *Significant after Holm correction.

TABLE I
MACRO-AVERAGED RESULTS ($n = 30$ PAIRED PRODUCTS). Δ = VIGALT − ASSETS24, IN PP (EFFICIENCY: RELEVANT NOVEL CLAIMS PER 100 WORDS).

Metric	ViGALT	ASSETS24	Δ [95% CI]	p_{Holm}	r
Relevancy ↑	**90.10**	48.12	+41.99 [30.9, 53.2]	<0.001*	0.99
Objectivity ↑	**97.71**	92.99	+4.72 [1.2, 8.4]	0.069	0.67
Rel. Novelty ↑	**36.12**	22.29	+13.84 [3.1, 24.2]	0.061	0.53
Avoid. Red. ↓	**17.76**	20.55	−2.80 [−9.2, 3.3]	0.828	0.14
Efficiency ↑	**5.94**	4.93	+1.01 [−1.0, 3.0]	0.828	0.18

*Significant at $\alpha = 0.05$ after Holm correction ($p_{\text{Holm}} = 4.6 \times 10^{-5}$). r: matched-pairs rank-biserial correlation. Bold: better value.

Objectivity (+4.72 pp, $p = 0.023$) and **Relevant Novelty** (+13.84 pp, $p = 0.015$) have confidence intervals that exclude zero and medium-to-large effect sizes ($r = 0.67$ and 0.53), but they do not survive Holm correction ($p_{\text{Holm}} = 0.069$ and 0.061). We therefore report them as consistent trends. Relevant Novelty favored ViGALT on 20 products versus 7. **Avoidable Redundancy** (17.76% vs. 20.55%) and **Information Efficiency** (5.94 vs. 4.93 relevant novel claims per 100 words) favor ViGALT on average, but their differences are small relative to between-image variation ($p_{\text{Holm}} = 0.83$). ViGALT's higher rate of *unavoidable* repetition (41.25% vs. 19.35%) is intended: it reflects anchor phrases that make new visual relations understandable and is not penalized.

Fig. 3 illustrates the relevancy gap: ASSETS24 spends much of its description on the model's accessories and bottoms and on the nursery's decor, whereas ViGALT describes the T-shirt's cut and chest graphic and the bed frame itself.

C. Fusion-Stage Validator Ablation

To isolate each validator, we re-ran only the fuser on the 30 products with one validator channel removed (`no_completeness`, `no_accuracy`, `no_redundancy`; 90 runs), holding the atomic claims, validator outputs, image, and DOM fixed. The reference here is full ViGALT, not ASSETS24. Table II summarizes the structural outcomes.



(a) Clothing

ASSETS24: A woman wears a black V-neck T-shirt with a white heart outline, layered necklaces, and bracelets, paired with distressed blue denim shorts. A patterned shoulder bag hangs at her side.

ViGALT: A T-shirt is shown worn on a person. It has a V-neckline, a straight body extending to the upper hips, and short sleeves ending above the elbows, with an irregular white outlined heart on the upper right chest.



(b) Furniture

ASSETS24: Bright nursery with a white house-style crib, pink-and-white bedding, plush toys, and string lights. A plush lion hangs on the wall, while a crocheted toy lies on the wooden floor beside an animal-patterned rug. A rounded inset in the upper left shows a white crib mattress. White walls and pale curtains complete the room.

ViGALT: A white-painted bed frame has a peaked roof-shaped frame with four corner posts. Raised guard panels made of vertical slats run along the sides and foot end, with a solid back panel behind the low-set sleeping surface and an opening at the front. The sleeping surface contains a white mattress with a lightly quilted cover.

Fig. 3. Qualitative comparison. Red marks purchase-irrelevant detail in ASSETS24; ViGALT text is unmarked.

TABLE II
FUSION-STAGE ABLATION: PRIMARY STRUCTURAL OUTCOMES ($n = 30$).

Validator removed	Primary outcome	Value
Completeness	Validator-added claims lost	48/48 (100%)
Completeness	Mean Δ claims / words	−1.37 / −9.47
Accuracy	Products with corrections/deletions	7/30
Redundancy	Mean Δ avoidable redundancy rate	+0.048

Completeness has the broadest effect. It added at least one claim for 27 of 30 products (48 claims in total, 1.60 per product). Without it, all 48 claims disappear, and outputs lose 1.37 claims and 9.47 words on average; for example, "short sleeves and a crew neckline" drop out of a T-shirt description. *Accuracy* interventions are sparse but decisive: 8 corrections and 1 deletion across 7 products. For instance, it corrects "black contrasting collar" to "gray collar with a black inner collar" and removes the unsupported claim "a sofa bed is arranged as a single bed". Without the validator, these errors return. *Redundancy* has a moderate effect: without it, more avoidable DOM repetition is retained in 14 of 30 products. Ablated variants score higher on post-hoc efficiency because validators deliberately add coverage and corrected wording. This trade-off is expected and does not mean that removing validators improves ViGALT.

D. Discussion

The pattern of results matches the design. Relevancy is where the two pipelines differ most by construction: ASSETS24 is asked to describe the image in context, whereas ViGALT is asked to describe the *product* in context, so scene elements that ASSETS24 reports correctly are counted as irrelevant here. Efficiency is not significant for a related reason: the completeness validator deliberately adds coverage, which lengthens ViGALT texts, and per-image efficiency varies widely with product complexity (ViGALT was higher on

13 products and ASSETS24 on 14). The ablation also points to a concrete improvement. Removing completeness at fusion lowers the post-hoc avoidable-redundancy rate from 0.178 to 0.045, which indicates that many completeness additions (e.g., sleeve length) are judged to repeat attributes that are already on the page. Tightening the completeness validator's DOM check is therefore the most direct way to improve ViGALT's redundancy score without sacrificing coverage.

V. Conclusion and Future Work

We presented ViGALT, a gap-aware alt-text pipeline that treats the product page as known context and uses claim-level validation to write concise, purchase-relevant descriptions for BLV shoppers. With the underlying model held constant, ViGALT produced significantly more relevant alt text than a strong context-aware baseline, with consistent trends in objectivity and relevant novelty. The ablation shows that each validator contributes something distinct: coverage from completeness, factual correction from accuracy, and less DOM repetition from redundancy. These findings are preliminary. The test set covers 30 images from one retailer and two product families, and all judgments are automated and rubric-based rather than drawn from BLV users. Future work will (i) run a user study with BLV participants to test whether higher rubric scores translate into better shopping decisions, (ii) add an independent factuality audit of final outputs, and (iii) extend ViGALT to more product categories and retailers.

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